

PRABAL PANDEY

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SUMMARY

Senior Software Engineer with **4+ years** building **Full Stack applications using C#/.NET 8, ASP.NET Core Web APIs, microservices, and Angular** for enterprise video analytics platforms deployed across **200+ servers and 1,000+ cameras**. Designs distributed systems using **RabbitMQ** and **PostgreSQL/TimescaleDB**, applying Clean Architecture and SOLID. Delivered **20x higher message throughput** and **12x faster exports**; leads code reviews, CI/CD, releases, and mentors **4 engineers**.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL (T-SQL, PL/pgSQL)

Backend: .NET 8, .NET Core, .NET Framework, ASP.NET Core Web API, REST APIs, Entity Framework Core

Architecture: Microservices, Event-Driven, Distributed Systems, Clean Architecture, SOLID, Design Patterns

Databases: PostgreSQL, Microsoft SQL Server, MySQL, TimescaleDB, pgvector, Milvus (vector search)

Messaging and Real-time: RabbitMQ, MassTransit, MQTT, SignalR, WebSockets, WebRTC, TCP/IP

Frontend: Angular 18, TypeScript, Kendo UI, GridStack, HTML5, CSS3

Cloud and DevOps: Docker, CI/CD, GitHub Actions, Git, Linux, Grafana, AWS (EC2, S3)

Testing and Practices: xUnit, Unit Testing, Architecture Tests, Code Review, Agile/Scrum

Domain: Video Surveillance, VMS, Video Analytics, ANPR, Face Recognition, ITMS, ICC, RTSP, Real-time Systems

PROFESSIONAL EXPERIENCE

I2V Systems Private Limited

Gurugram, India

Senior Software Engineer

Apr 2026 - Present

- Led the **Clean Architecture** migration of the **ASP.NET Core** analytics monolith into **7 modules** (Events, FRS, Actions, Dashboard, Report, Logs, Schedule) with per-module **EF Core** migrations and architecture tests.
- Optimized **PostgreSQL** report exports from about **12 minutes to under 1 minute** on **100K+ records (12x faster)** using streaming reads and bounded parallelism across **10 report types**.
- Own **cross-platform releases** (Windows, Linux, Docker) from **GitHub Actions CI/CD** pipelines to public-sector sign-off; lead **Grafana**-based incident triage and code reviews; mentor **4 junior engineers** in Agile/Scrum.

Software Engineer

Aug 2022 - Apr 2026

- Rebuilt the **TimescaleDB** event store (**40–50M events**) by merging per-type tables into one hypertable with **JSONB** and **BRIN** indexes, cutting search time from **30s to under 3s (10x)**.
- Scaled **RabbitMQ** event processing from **20–30 to 500+ events/sec (20x)** and eliminated daily broker restarts by splitting a mixed queue into per-event queues with independent consumers and dead-letter queues.
- Integrated third-party systems (**DigiFort VMS**, **MQTT** alerting) through a composable action pipeline that attaches **6 action types** to any event, so new integrations ship as configuration instead of code.
- Built the dashboard platform and **Angular 12 to 18** applications (live video walls, **4 real-time dashboards**) on a **PostgreSQL** query engine with group-by and joins, published **4 npm libraries**, and upgraded **.NET 6 to .NET 8**.
- Shipped **Milvus** vector search into the **Face Recognition System (FRS)** behind a pluggable store interchangeable with **pgvector**, choosing **IVF SQ8** over **HNSW** for a **75% smaller** index.

PROJECTS

PhotoGrouper | Offline face grouping | github.com/kingpingx/PhotoGroup

2026 - Present

- Building a **.NET 8/Avalonia** desktop app that detects, aligns and embeds faces with **ONNX Runtime** (YuNet, ArcFace) and clusters them into people (**Chinese Whispers**), resumable per stage on **SQLite**.

ThePlayer | RTSP and file playback in Chrome | github.com/kingpingx/ThePlayer

2026 - Present

- Building a **.NET 8/FFmpeg/MediaMTX** player with 3 decode paths: **H.265 to H.264** transcode, original bytes over **WebSocket/WebCodecs**, and **WebRTC**, all from one shared FFmpeg process.

EDUCATION AND ACHIEVEMENTS

B.Tech in Software Engineering, Maharshi Dayanand University

Aug 2019 - Jun 2023

Top 8, Microsoft Imagine Cup India Pre-Finals (Feb 2022)